



Maven Toys Retail Sales & Inventory Optimization

Detailed Analysis Report

Sales period: January 2022 – September 2023

SQL • BigQuery • Looker Studio • Business Analysis

Prepared By

Vijay Kumar L

Executive Summary

This case study evaluates Maven Toys across sales performance, product economics, geography, store performance, pricing, inventory risk and seasonality. The goal is to demonstrate an end-to-end analytical workflow: validate the data, analyze it with SQL, communicate findings through Looker Studio, and translate evidence into practical business recommendations.

Area	Key finding
Business performance	\$14.44M revenue, 1.09M units sold, \$4.01M gross profit and 27.79% gross margin.
Product concentration	The first 15 products reach 80.08% cumulative revenue; Gamer Headphones is the 15th product and crosses the 80% threshold.
Category economics	Toys contributes 35.26% of revenue but has the lowest margin at 21.20%. Electronics has the highest margin at 44.57% and about \$1.00M gross profit.
Geography	Ciudad de Mexico leads city revenue at \$1.65M (11.42%); the top five cities contribute 41.58% of company revenue.
Store performance	Ciudad de Mexico 2 is the top individual store at about \$0.55M revenue.
Inventory risk	Guanajuato 1 has the highest reported high-risk share at 58.62%; Xalapa 1 is 48.28% and Oaxaca 1 is 44.83%.
Seasonality	2022 revenue strengthened into April, softened through August, then recovered. 2023 data ends in September, so October–December comparisons are incomplete.

Management Priorities

- Protect availability of the highest-contributing products, particularly the top 15 products that collectively reach approximately 80% of revenue.
- Investigate whether Electronics can be scaled because it combines relatively strong revenue with the highest gross margin.
- Review the Toys category for pricing, product mix and cost opportunities because it produces the largest revenue contribution but the lowest category margin.
- Prioritize inventory review in stores with high-risk shares, using the risk indicator as a directional planning tool rather than proof of historical stockouts.
- Use city and store-level performance to prioritize commercial activity and benchmark lower-performing branches against leading stores in the same city.
- Use monthly and seasonal patterns to support inventory and promotion planning while recognizing incomplete 2023 coverage.

1. Business Context & Objective

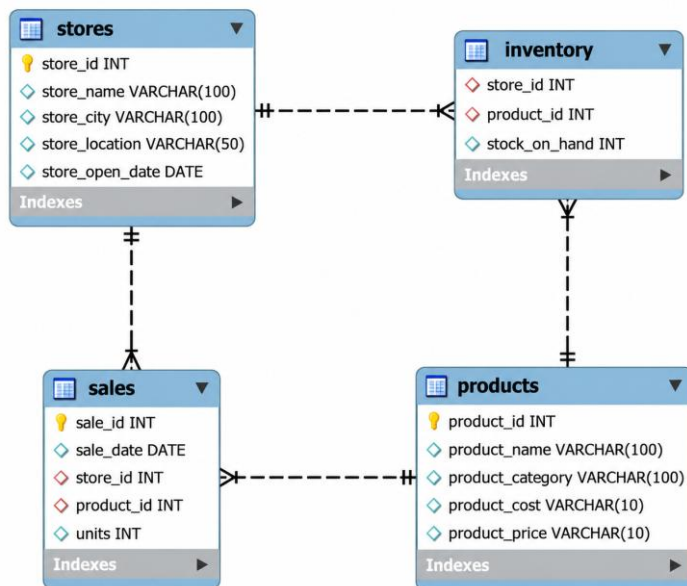
Maven Toys is analyzed as a retail network using transaction, product, store and inventory data. The analysis answers the original case questions while extending them into a portfolio-grade business analysis. The final report intentionally focuses on business evidence and interpretation; detailed SQL is maintained separately in GitHub.

2. Data & Methodology

The analytical model uses four related tables: sales, products, stores and inventory.

- Revenue = units sold × product price
- Gross profit = units sold × (product price – product cost)
- Gross margin = gross profit ÷ revenue
- Inventory risk = current stock compared with historical average monthly unit sales

ER Diagram



Data Quality Summary

Dataset scale: 50 stores, 35 products, 829,262 sales transactions and 1,593 inventory records.

Sales period: 2022-01-01 to 2023-09-30, covering 638 distinct selling days.

Referential integrity: No invalid store or product references were found in sales.

Duplicates: Inventory contains 7 duplicate full records; stores, products and sales show no duplicate full records in the source analysis.

Inventory completeness: 1,750 theoretical store-product combinations exist; 157 have no inventory record.

Inventory limitation: Inventory is a current snapshot, not a time-stamped history. Risk analysis therefore compares current stock with historical sales velocity.

3. Overall Sales Performance

The business generated \$14.44M revenue from 1,090,565 units, with \$10.43M product cost and \$4.01M gross profit, resulting in a 27.79% gross margin.

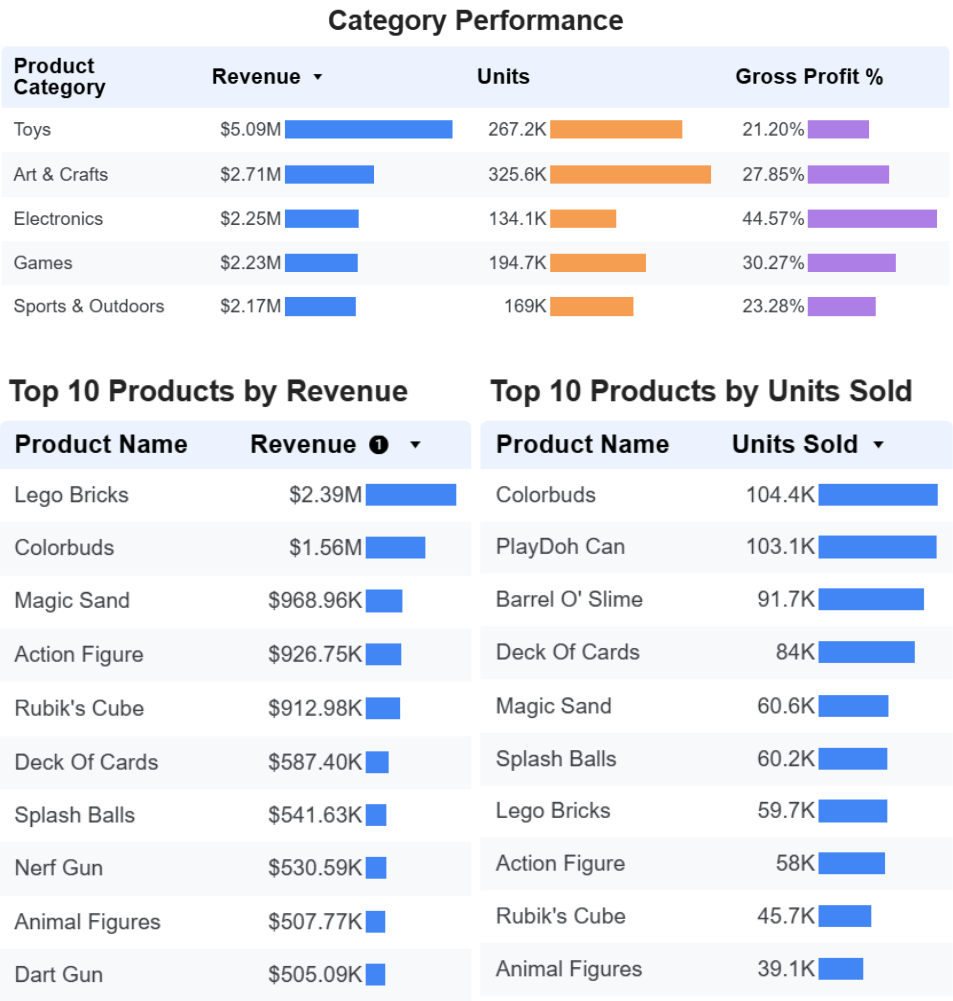
- The portfolio should evaluate revenue quality as well as revenue volume because product and category margins vary materially.



4. Product & Category Performance

Lego Bricks leads revenue at \$2.39M, while Colorbuds leads unit volume at 104,368 units. This demonstrates why revenue, volume and margin should be considered together.

- Toys is the largest category at \$5.09M and 35.26% of company revenue, but its 21.20% margin is the lowest category margin.
- Electronics generates \$2.25M revenue, contributes 15.55% of revenue, and produces the highest category margin at 44.57% and about \$1.00M gross profit.

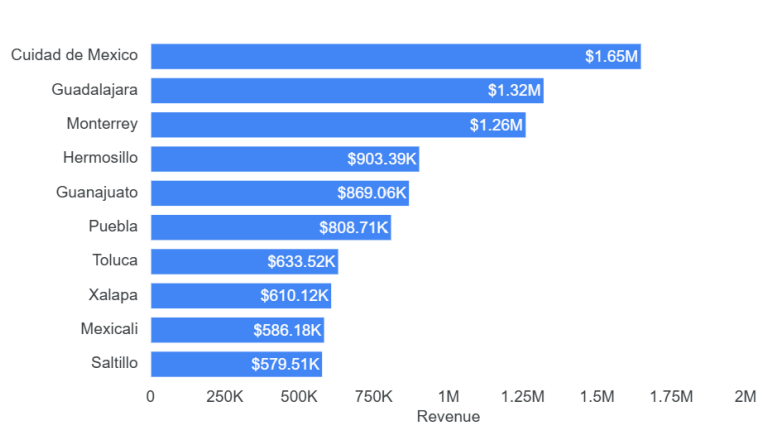


5. Geography & Store Performance

Ciudad de Mexico leads city revenue at \$1.65M, followed by Guadalajara at \$1.32M and Monterrey at \$1.26M. The top five cities contribute 41.58% of revenue.

- Ciudad de Mexico 2 is the leading store at about \$0.55M revenue; Guadalajara 4 follows at about \$0.35M.
- Downtown stores generate \$8.22M across 29 stores and Commercial stores generate \$3.28M across 12 stores; total revenue alone should not be treated as store-level productivity.

Revenue by City



Store Performance by City

Store Name	Store City	Units Sold	Revenue	Gross Profit %
Maven Toys Ciudad de Mexico 2	Ciudad de Mexico	42.76K	\$554.55K	30.63%
Maven Toys Guadalajara 4	Guadalajara	24.14K	\$348.47K	29.32%
Maven Toys Ciudad de Mexico 4	Ciudad de Mexico	22.73K	\$323.96K	27.90%
Maven Toys Monterrey 4	Monterrey	22.45K	\$325.07K	24.41%
Maven Toys Hermosillo 2	Hermosillo	23.50K	\$323.43K	27.21%
Maven Toys Xalapa 2	Xalapa	25.30K	\$344.31K	25.74%
Maven Toys Guanajuato 3	Guanajuato	18.44K	\$276.21K	24.44%

Location Type Performance

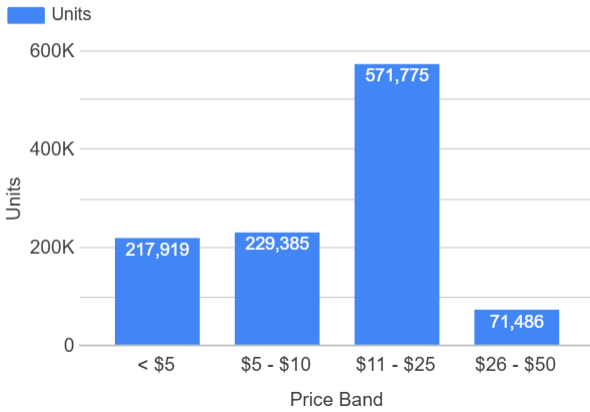
Store Location	Revenue	Units	Gross Profit %
Downtown	\$8.22M	628.1K	27.36%
Commercial	\$3.28M	240.6K	28.27%
Residential	\$1.66M	125.4K	27.80%
Airport	\$1.29M	96.4K	29.31%

6. Product Distribution & Pricing

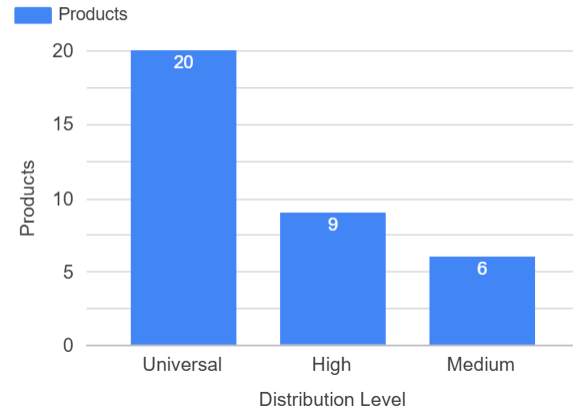
Current inventory records show 20 Universal products, 9 High and 6 Medium, with no Low products. Distribution reflects recorded inventory presence, not historical availability.

- Price does not show a simple one-direction relationship with unit volume: low-priced products can sell heavily, while premium products such as Lego Bricks can also generate substantial sales.
- Same-price examples include Classic Dominoes/Jenga at \$9.99, Monopoly/Rubik's Cube at \$19.99, and Animal Figures/Teddy Bear at \$12.99.

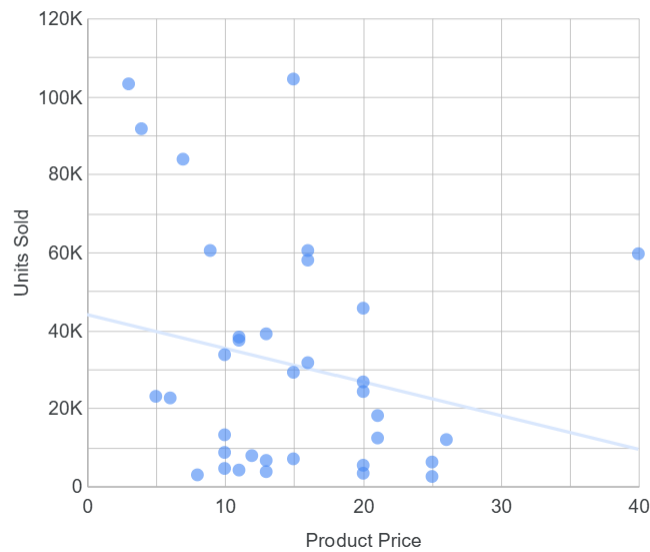
Units Sold by Price Band



Products by Distribution Level



Product Price Vs Sales Volume



7. Inventory Risk & Optimization

The inventory-risk model compares current stock with historical average monthly sales and classifies records as High Risk, Potential Reorder, Normal or No Historical Sales.

- Guanajuato 1 has the highest reported high-risk share at 58.62%; Xalapa 1 is 48.28%; Oaxaca 1 is 44.83%.
- Three inventory combinations have stock but no historical sales: PlayDoh Playset at Guanajuato 1 and Oaxaca 1, and Teddy Bear at Xalapa 1.
- These results should trigger review rather than automatic action because the inventory table is not time-stamped.

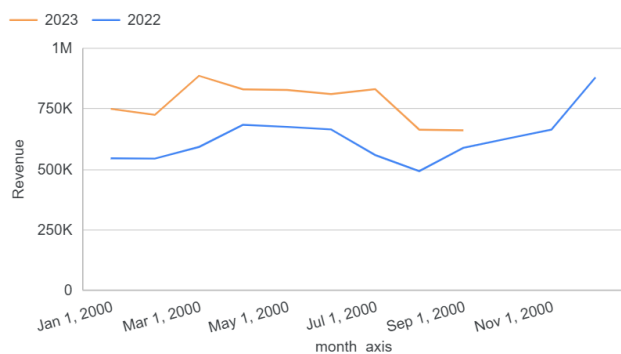
store_id	store_name	high_risk_products	potential_reorder_products	normal_products	no_historical_sales_products	total_inventory_products	high_risk_percentage
14	Maven Toys Guanajuato 1	17	6	5	1	29	58.62%
27	Maven Toys Oaxaca 1	13	8	7	1	29	44.83%
29	Maven Toys Xalapa 1	14	9	5	1	29	48.28%
24	Maven Toys Aguascalientes 1	13	9	13	0	35	37.14%
10	Maven Toys Campeche 1	13	10	12	0	35	37.14%
26	Maven Toys Campeche 2	14	5	16	0	35	40.00%
12	Maven Toys Chetumal 1	8	8	19	0	35	22.86%
23	Maven Toys Chihuahua 1	11	10	14	0	35	31.43%
38	Maven Toys Chihuahua 2	11	7	17	0	35	31.43%
35	Maven Toys Chilpancingo 1	10	6	13	0	29	34.48%

8. Time & Seasonality

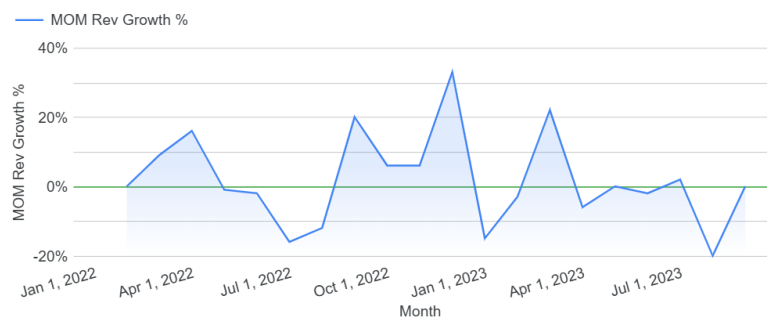
Monthly revenue was approximately \$0.54M in January 2022, peaked at \$0.68M in April, fell to \$0.49M in August and recovered to \$0.62M in October.

- The largest positive month-over-month movement in the available 2022 sequence is August to September at approximately +19.67%.
- 2023 data ends in September, so October–December year-over-year comparisons are unavailable.

Monthly Revenue — Year-over-Year Comparison



MOM Rev Growth % by Month

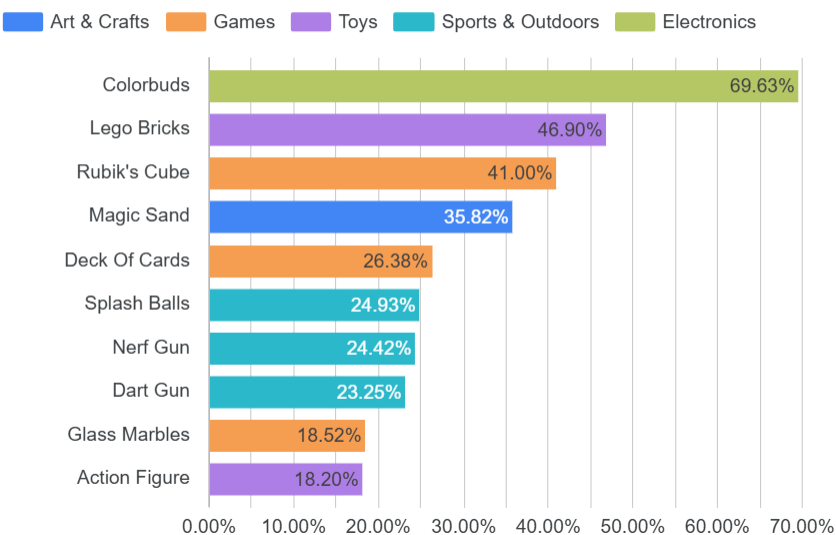


9. Advanced Analysis

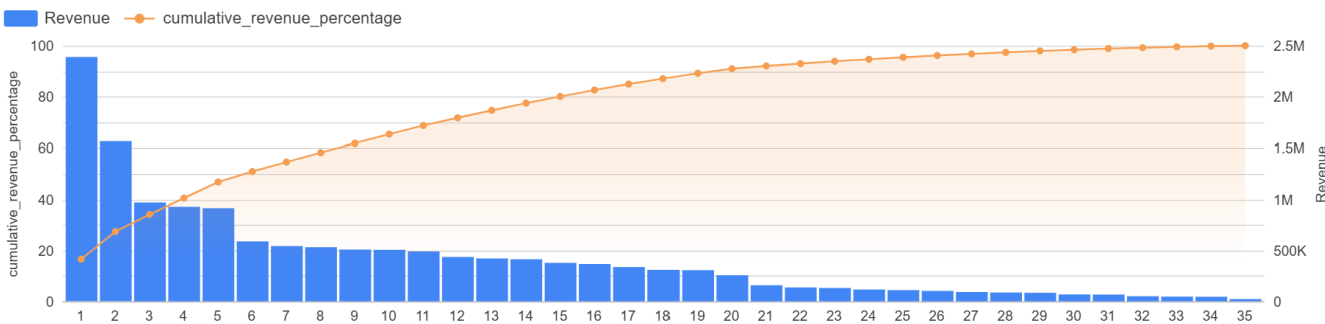
Window functions and contribution analysis add decision value beyond basic aggregation.

- Colorbuds contributes 69.63% of Electronics revenue; Magic Sand contributes 35.82% of Art & Crafts revenue.
- Pareto analysis shows the first five products account for 46.81% of revenue and the first 15 reach 80.08%. Gamer Headphones is the 15th product and the first point at which cumulative revenue exceeds 80%.

Product Revenue Contribution Within Category



Product Revenue Pareto Analysis



10. Management Recommendations

1. Protect high-contribution products with stronger availability and replenishment attention.
2. Investigate Electronics as a potential margin-led growth opportunity.
3. Review Toys pricing, product mix and cost structure to improve margin without damaging volume.
4. Use store and city benchmarks to target underperforming branches with localized actions.
5. Prioritize review of high-risk inventory locations and investigate records with no historical sales.
6. Align inventory and promotional planning with observed monthly/seasonal demand patterns.

11. Data Limitations & Analytical Caveats

- Missing inventory records should not automatically be interpreted as zero stock.
- Inventory risk is an indicator based on current stock and historical sales velocity, not a historical stockout model.
- Product distribution is based on inventory records and should not be described as historical availability.
- Inventory duplicates should be considered when calculating distribution or risk counts.
- 2023 covers only January through September, so late-year seasonality is incomplete.
- Month-over-month growth should be displayed as percentage values such as +19.67%, not the underlying decimal fraction.

12. Portfolio Deliverables

- GitHub: SQL scripts organized by analytical phase.
- Looker Studio: interactive dashboards for sales, geography/stores, products/pricing, time/seasonality and advanced analysis.
- Report: business-facing findings, interpretation and recommendations.
- README: project context, business questions, data model, tools, dashboard link, key insights and repository structure.

Appendix — Analytical Question Map

Theme	Questions	Coverage
Data understanding	Q1–Q5	Profiling, dates, integrity, duplicates, inventory completeness
Core sales	Q6–Q10	Overall performance, top products, categories, stores
Geography & stores	Q11–Q15	City, location type, store age, store contribution, concentration
Products & pricing	Q16–Q22	Distribution, price bands, price-volume, same-price, margin, revenue/profit
Inventory optimization	Q23–Q24	Inventory risk and high-risk store concentration
Time & seasonality	Q25–Q27	Monthly sales, MoM growth, cumulative revenue
Advanced SQL	Q28–Q30	Store ranking, category contribution, Pareto concentration

Conclusion

Maven Toys demonstrates an end-to-end retail analytics case in which SQL is the analytical engine, Looker Studio is the communication layer, and business recommendations are the final deliverable. The strongest portfolio story is the ability to move from a business question to reliable data, clear evidence, an interpretable insight and a practical management action.